Bonus Exam 3	**SOLUTION**	29.09.2025
Advanced Database Systems (**184.780**)		Duration: 75 minutes

Do not hand in this or any of the following question sheets.
Nothing on them will be graded.

Problem 1: (4)

These are true multiple choice questions. Any number of answers (between 0 and all) could be correct. 1 credit for each question.

Question 1: Consider the Two-Phase-Commit Protocol in case of a transaction T with 1 coordinator and 3 subordinates. In which of the following scenarios is it guaranteed that transaction T will eventually commit? Here we assume that every process that crashes will eventually fully recover. However, we may not assume that the recovery happens fast enough to stay within specified time-outs of communication partners.

- **(a)** One of the subordinates receives a commit message for transaction T and crashes before it can write a commit log record to disk. We have no information on the other two subordinates.
- **(b)** All three subordinates receive a commit message for transaction T. One of the subordinates crashes before it can write a commit log record to disk whereas the other two subordinates manage to write the commit log record to disk.
- **(c)** One of the subordinates receives a prepare message for transaction T and crashes before it can send a yes message. We have no information on the other two subordinates.
- **(d)** All three subordinates receive a prepare message for transaction T. One of the subordinates crashes before it can send a yes message whereas the other two subordinates manage to send a yes message.

(Correct: a,b)

Explanation:
Answers a and b are correct.
ad answers a and b: before sending a commit message, the coordinator has written a commit log to disk. Hence, it considers the transaction as successful. Subordinates that crash at this point are guaranteed to have a prepare log entry on disk. Hence, after recovery, they will ask the coordinator about the status of the transaction.
ad answers c and d: if the coordinator does not receive a yes from all subordinates then, after a specified time-out, it will decide to abort the transaction.
For further information, see the slides III.1, pp. 38 - 40.

Question 2: Consider a scenario where 3 processes a, b, c are synchronized via vector clocks. We denote clock values as triples (v_1, v_2, v_3). If such a triple is the current clock value at one of the processes, then (from the point of view of this process), process a has value v_1, process b has value v_2, and process c has value v_3. Which of the following statements concerning possible values of vector clocks are correct?

- **(a)** There exists a message flow between the 3 processes a, b, c, such that process a has the clock value $(3, 3, x)$ for some value x and process b has the clock value $(3, 3, y)$ for some value y.
- **(b)** There exists a message flow between the 3 processes a, b, c, such that process a has the clock value $(3, 3, x)$ for some value x and process b has the clock value $(3, 4, y)$ for some value y.
- **(c)** There exists a message flow between the 3 processes a, b, c, such that process a has the clock value $(4, 3, x)$ for some value x and process b has the clock value $(3, 3, y)$ for some value y.
- **(d)** There exists a message flow between the 3 processes a, b, c, such that process a has the clock value $(3, 3, x)$ for some value x and process b has the clock value $(4, 3, y)$ for some value y.

(Correct: a,b,c)

Explanation:
Answers a, b, and c are correct. See the Slides III.2, p.47.
ad answers a,b,c: for instance, there could be 3 messages sent between processes a and b; additionally, there is 1 message sent between process c and b (in case of answer b) or between c and a (in answer c).
ad answer d: it cannot happen that process b has a higher value for process a than a itself.

Question 3: Consider various encoding techniques for column stores. Of course, their concrete behaviour strongly depends on the concrete data that is being encoded. Which of the following statements are correct?

(a) It may happen that bit vector encoding requires more space than run-length encoding.

(b) It may happen that run-length encoding requires more space than bit vector encoding.

(c) It may happen that run-length encoding requires more space than dictionary encoding.

(d) It may happen that dictionary encoding requires more space than run-length encoding.

(Correct: a,b,c,d)

Explanation:
Answers a,b,c, and d are correct: see Slides III.5, pp. 16-18.
ad answer a,d: consider, e.g., the extreme case of a large number of tuples with a single value in a column.
ad answer b,c: e.g.: 2 constantly alternating distinct values.

Question 4: Which of the following read/write quorums are allowed in the Quorum-Consensus protocol?

(a) In case of replication factor $N = 15$: $w = 8$ and $r = 7$.

(b) In case of replication factor $N = 15$: $w = 7$ and $r = 8$.

(c) In case of replication factor $N = 15$: $w = 9$ and $r = 7$.

(d) In case of replication factor $N = 15$: $w = 7$ and $r = 9$.

(Correct: c)

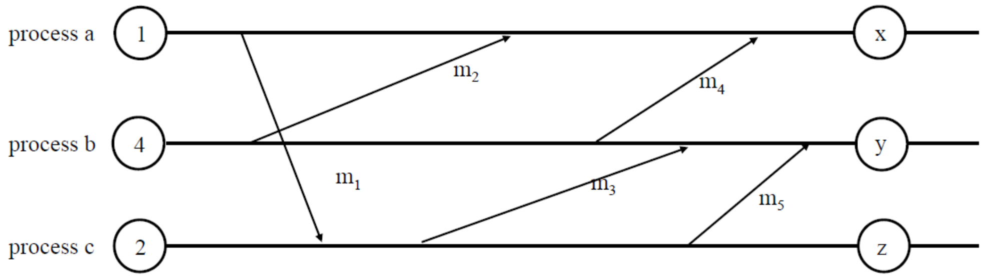
Explanation:
Answer c is correct. See Slides III.2, p.41.

Problem 2:

(4)

Below are true multiple choice questions. Any number of answers (between 0 and all) could be correct. 1 credit for each question.

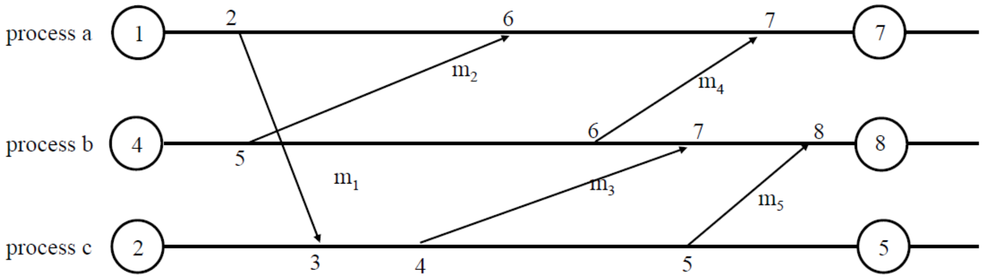
Scalar Clocks: [1 credit]
Consider a scenario with 3 processes. We observe these processes in a small time window. The processes have already started. Therefore, at the beginning of the time window, the clock counts are not 0. Instead they are 1, 4, and 2, respectively. Within this time window, 5 messages $m_1, \ldots, m_5$ are sent between processes. Each of the circles labelled x, y, z, respectively, stands for the clock value of one of the processes at the end of the time window. Your task is to determine these clock values.



Question 5: Which of the following clock values are correct:

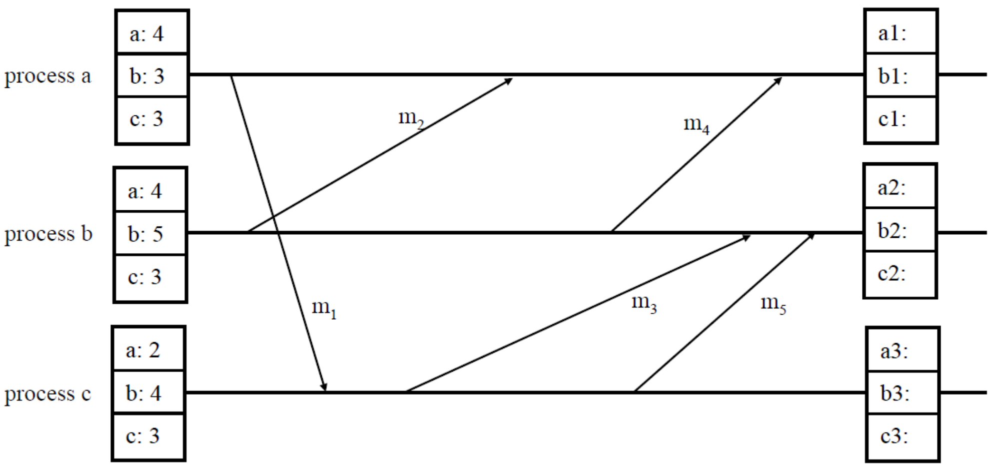
(a) $x = 7$ **(b)** $y = 7$ **(c)** $z = 4$ **(Correct: a)**

Explanation:
Answer a is correct. See the Slides III.2, p.44 and the sample solution in TUWEL.



Vector Clocks: [3 credits]
Consider a scenario with 3 processes. We observe these processes in a small time window. The processes have already started. Therefore, at the beginning of the time window, the vector clocks have values different from (0,0,0). Instead, they are (4,3,3), (4,5,3), and (2,4,3), respectively. Within this time window, 5 messages $m_1, \ldots, m_5$ are sent between processes.

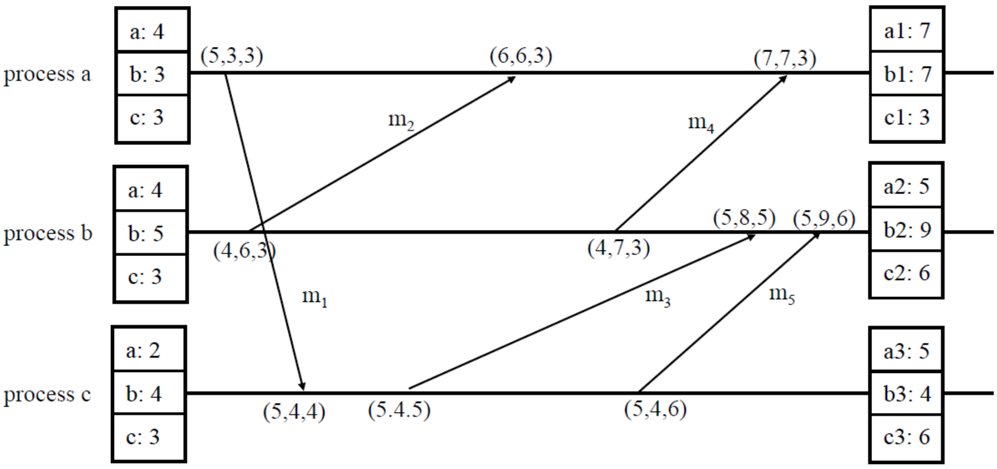
At the end of the window, we want to know the final vector clock values at each process, represented by the labels $(a1, b1, c1)$, $(a2, b2, c2)$, and $(a3, b3, c3)$, respectively. Your task is to determine the correct values of these clocks at the end of the time window.



Question 6: Which of the following clock values are correct for process a:

(a) $a1 = 7$ **(b)** $b1 = 7$ **(c)** $c1 = 4$ **(Correct: a,b)**

Explanation:
Answers a and b are correct. See the Slides III.2, p.47 and the sample solution in TUWEL.



Question 7: Which of the following clock values are correct for process b:

(a) $a2 = 7$ **(b)** $b2 = 7$ **(c)** $c2 = 6$ **(Correct: c)**

Explanation:
Answer c is correct. See the Slides III.2, p.47 and the sample solution in TUWEL.

Question 8: Which of the following clock values are correct for process c:

(a) $a3 = 5$ **(b)** $b3 = 7$ **(c)** $c3 = 6$ **(Correct: a,c)**

Explanation:
Answers a and c are correct. See the Slides III.2, p.47 and the sample solution in TUWEL.

Problem 3: (3)

Below are true multiple choice questions. Any number of answers (between 0 and all) could be correct. 1 credit for each question.

Consider a relational database with information on students and lectures as well as information as to which student has already taken the exam of which lecture (together with the date of the exam). Suppose that the (very simplified) schema is as follows:

Students (s_id, name, studies), Exams (student, lecture, date) Lectures (l_id, title, prof)

where Exams.student and Exams.lecture are foreign keys into the Students and Lectures table, respectively. Suppose that these tables have the following content:

Students

s_id	name	studies
S1	alice	645
S2	bob	937
S3	charlie	357

Exams

student	lecture	date
S1	L1	2024
S2	L1	2025
S1	L3	2023
S3	L3	2024

Lectures

l_id	title	prof
L1	ADBS	carol
L2	ThInf	david
L3	EP2	emma

Below you find three realizations (with gaps) of this contents as databases in a document store.

Database DB1

Exams: [
{ "student": "S1",
"name": "alice",
"studies": 645,
"lecture": "L1",
"title": "ADBS",
"prof": "carol",
"date": 2024 },

{ Q9a ,
"lecture": "L1",
"title": "ADBS",
"prof": "carol",
"date": 2025 },

{ "student": "S1",
"name": "alice",
"studies": 645,
Q9b },

{ Q9c ,
"lecture": "L3",
"title": "EP2",
"prof": "emma",
"date": 2024 }]

Database DB2

Students: [
{ "s_id": "S1",
"name": "alice",
"studies": 645,
"exams":
[Q10a ,
{ "l_id": "L3",
"title": "EP2",
"prof": "emma",
"date": 2023 }]
},

{ "s_id": "S2",
"title": "bob",
"studies": 937,
"exams":
[Q10b]
},

{ "s_id": "S3",
"title": "charlie",
"studies": 357,
"exams":
[Q10c]
}]

Database DB3

Students:
[{ "s_id": "S1",
"name": "alice",
"studies": 645,
"exams": Q11a
},

{ "s_id": "S2",
"name": "bob",
"studies": 937,
"exams": ["L1"] } ,

{ "s_id": "S3",
"name": "charlie",
"studies": 357,
"exams": Q11b
}]

Lectures:
[{ "l_id": "L1",
"title": "ADBS",
"prof": "carol",
"exams": Q11c
},

{ "l_id": "L2",
"title": "ThInf",
"prof": "david",
"exams": Q11d
},

{ "l_id": "L3",
"title": "EP2",
"prof": "emma",
"exams": ["S1", "S3"] }]

Question 9: Suppose that **Database DB1** has been designed in such a way that the entire information on Students, Exams, and Lectures is contained in a single collection. More specifically, the Students and Lectures collections are denormalized into Exams. Which of the following statements are correct if your task is to fill in the missing parts into Database DB1. In particular, you should replace the boxes Q9a, Q9b, and Q9c by the correct code snippets.

(a) $\boxed{\text{Q9a}}$ has to be replaced by the following code snippet:
"student": "S3", "name": "charlie", "studies": 357

(b) $\boxed{\text{Q9b}}$ has to be replaced by the following code snippet:
"lecture": "L3", "title": "EP2", "prof": "emma", "date": 2023

(c) $\boxed{\text{Q9c}}$ has to be replaced by the following code snippet:
"student": "S3", "name": "charlie", "studies": 357

(d) The type of denormalization applied in Database DB1 is called "denormalizing into the one-side".

(Correct: b, c)

Explanation:
Answers b and c are correct. See Slides III.3, p.24/25 and the sample solution in TUWEL.

Question 10: Suppose that **Database DB2** has been designed in such a way that the Students collection contains the entire information on Lectures they have taken Exams in. Which of the following statements are correct if your task is to fill in the missing parts into Database DB2. In particular, you should replace the boxes Q10a, Q10b, and Q10c by the correct code snippets.

(a) $\boxed{\text{Q10a}}$ has to be replaced by the following code snippet:
{ "l_id": "L1", "title": "ADBS", "prof": "carol", "date": 2024 }

(b) $\boxed{\text{Q10b}}$ has to be replaced by the following code snippet:
{ "l_id": "L3", "title": "EP2", "prof": "emma", "date": 2025 }

(c) $\boxed{\text{Q10c}}$ has to be left empty (i.e., there is nothing missing here)

(d) One motivation for choosing this type of denormalization could be to speed up the following type of queries: for a given lecture, obtain detailed information on students having taken this exam.

(Correct: a)

Explanation:
Answer a is correct. See Slides III.3, p.24/25 and the sample solution in TUWEL.

Question 11: Suppose that **Database DB3** has been designed in such a way that the Students collection contains for each student the list of l_id's of the lectures whose exams she/he has already taken. Likewise, the Lectures collection contains for each lecture the list of s_id's of the students that have already taken that exam. Due to lack of space, we are not showing the Exams collection, which is of course also contained in Database DB3. Which of the following statements are correct if your task is to fill in the missing parts into Database DB3. In particular, you should replace the boxes Q11a, Q11b, Q11c, and Q11d by the correct code snippets.

(a) $\boxed{\text{Q11a}}$ has to be replaced by the following code snippet: ["L1", "L3"]

(b) $\boxed{\text{Q11b}}$ has to be replaced by the following code snippet: ["L1"]

(c) $\boxed{\text{Q11c}}$ has to be replaced by the following code snippet: ["S1", "S2"]

(d) $\boxed{\text{Q11d}}$ has to be left empty (i.e., there is nothing missing here)

(Correct: a,c)

Explanation:
Answer a, c is correct. Answer d is wrong since $\boxed{\text{Q11d}}$ has to be filled with [ ] and not left empty. See Slides III.3, p.24/25 and the sample solution in TUWEL.

Problem 4: (5)

These are true multiple choice questions. Any number of answers (between 0 and all) could be correct. 1 credit for each question.

(5.1) Evaluate the following Cypher query on the Database given on the next page:

```
MATCH (c1:City)-[:HAS_AIRPORT]->(a:Airport)-[f:FLIGHT]->(:Airport)<-[:HAS_AIRPORT]-(c2:City)
WHERE c1.name = 'Paris' AND f.cost < 170
RETURN DISTINCT c2.name
```

Question 12: Mark exactly those values that are returned by the query.

(a) Milan **(b)** Vienna **(c)** Zurich **(d)** Berlin **(e)** Paris **(f)** The query is not valid

(Correct: a, c, d)

(5.2) Evaluate the following Cypher query on the Database given on the next page:

```
MATCH p = allshortestPaths((a1:Airport)-[:FLIGHT*]->(a2:Airport {name : "MXP"}))
WHERE a1<>a2
RETURN a1.name ,length(p)
```

Question 13: Mark exactly those values that are returned by the query.

(a) TXL, 0 **(b)** ZRH, 1 **(c)** BER, 2 **(d)** BER, 3 **(e)** TXL, 2
(f) LIN, 1 **(g)** CDG, 1 **(h)** LIN, 3 **(i)** CDG, 2 **(j)** The query is not valid

(Correct: b, d, g)

(5.3) Evaluate the following Cypher query on the Database given on the next page:

```
MATCH (c1:City)-[:HAS_AIRPORT]->(a1:Airport)-[f:FLIGHT|TRAIN]->(a2:Airport)<-[:HAS_AIRPORT]-(c2:City)
WHERE c1.name = "Paris" AND c2.name = "Berlin"
RETURN a1.name, a2.name, f.cost
```

Question 14: Mark exactly those values that are returned by the query.

(a) ORY, BER, 75 **(b)** ORY, TXL, 60 **(c)** CDG, TXL, 80 **(d)** ORY, TXL, 40
(e) CDG, BER, 180 **(f)** The query is not valid

(Correct: b, c, e)

(5.4) Evaluate the following Cypher query on the Database given on the next page:

```
MATCH (a:Airport)-[f1:FLIGHT]->(), ()-[f2:FLIGHT]->(a)
WITH a, count(DISTINCT f1) AS outFlights, count(DISTINCT f2) AS inFlights
WHERE outFlights >= 2 OR inFlights >=2
RETURN a.name, outFlights, inFlights
```

Question 15: Mark exactly those values that are returned by the query.

(a) ORY, 2, 1 **(b)** ZRH, 1, 3 **(c)** VIE, 1, 1 **(d)** TXL, 1, 3 **(e)** MXP, 1, 2
(f) ZRH, 1, 2 **(g)** *The query is not valid*

(Correct: a, b, e)

(5.5) Which of the following Cypher queries correctly answers the question below?

"List the airports from which the VIE airport can be reached with at most one flight change."

Question 16: Mark the queries that return the correct result.

(a)
```
MATCH (a1:Airport)-[:FLIGHT*..2]->(a2:Airport{name:'VIE'})
RETURN DISTINCT a1.name
```

(b)
```
MATCH (a1:Airport)-[:FLIGHT*1..2]->(a2:Airport{name:'VIE'})
RETURN DISTINCT a1.name
```

(c)
```
MATCH p = (a1:Airport)-[f:FLIGHT*]->(a2:Airport{name:'VIE'})
WHERE length(p)<2
RETURN DISTINCT a1.name
```

(d)
```
MATCH (a1:Airport)-[:FLIGHT]->(a2:Airport{name:'VIE'})
MATCH (a1:Airport)-[:FLIGHT]->(:Airport)-[:FLIGHT]->(a2:Airport{name:'VIE'})
RETURN DISTINCT a1.name
```

(Correct: a, b)

Overall: 16 points

Graph DB Data Model

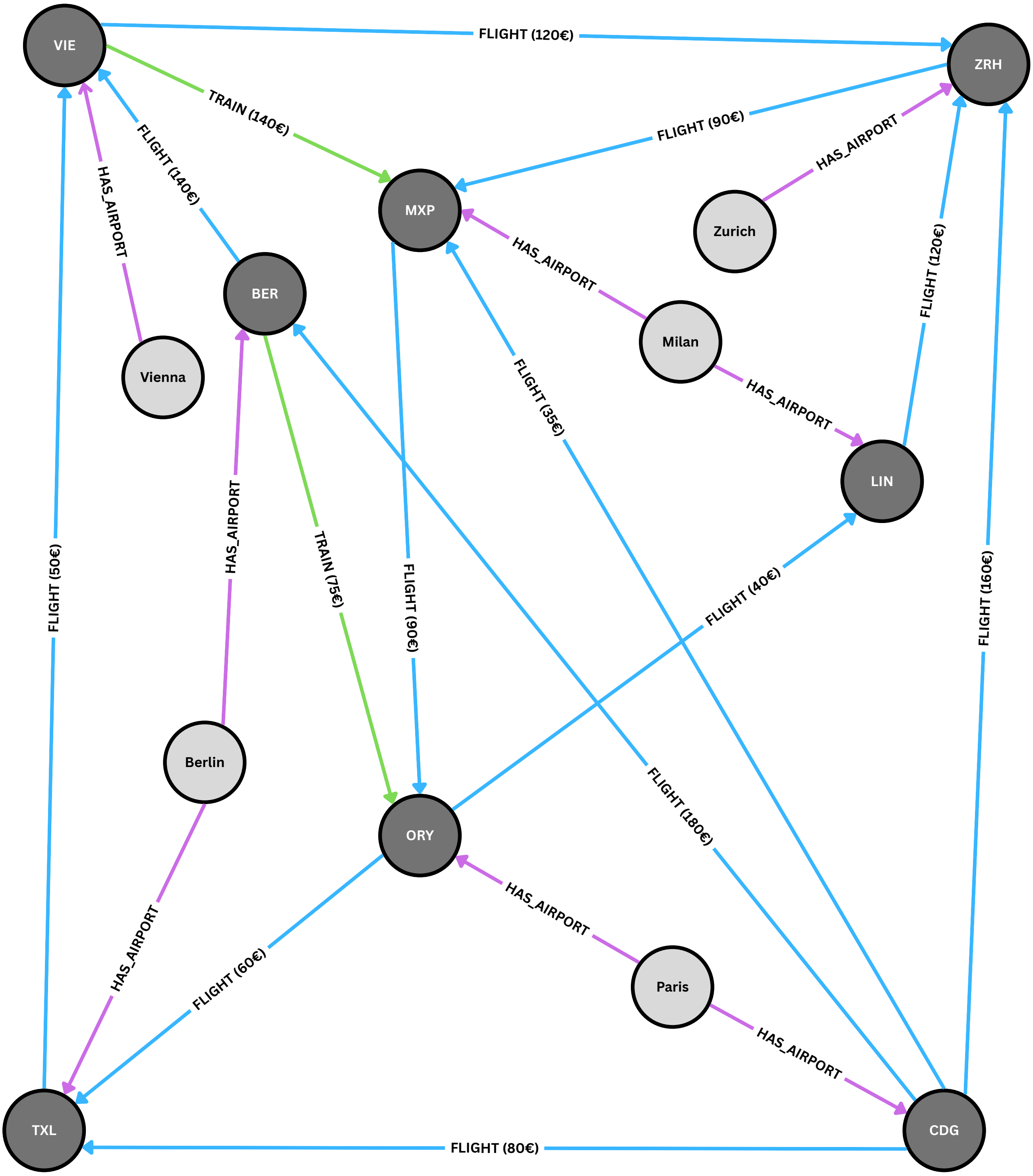
The graph contains nodes for cities with the label `:City` (light nodes), and nodes for airports with the label `:Airport` (dark nodes). A city can be connected to one or more airports using a relationship of type `:HAS_AIRPORT`.

A relationship of type `:FLIGHT` between two airports represents a direct flight. Each `:FLIGHT` relationship has the property `cost`, which is an integer that represents the cost of the flight in euros.

A relationship of type `:TRAIN` between two airports represents a connection by train. Each `:TRAIN` relationship also has a `cost` property, containing the cost of the train journey in euros.

`:FLIGHT` and `:TRAIN` relationships cannot connect a node to itself.

Graph Database



Problem 5: (4)

The questions below have exactly 1 correct answer. In the answer options a – e, you are presented 5 different values. Answer option f means, that none of the values presented in a – e is correct. The number of credits is given for each question separately.

Assume a distributed database with 3 sites and the relations `Rental` and `Car`. See the following tables for details of the scenario. You can assume that the relations are *non-fragmented* and all records (and attributes) are *fixed-size*. Note that the attributes shown are a subset of all attributes.

Relation	Site	# Records	Record Size (byte)
Rental	1	10 000	200
Car	2	100	500

Relation	Attribute	Size
Rental	rid	10
Rental	car	20
Rental	until	10
Car	cid	20
Car	type	40
Car	description	200

Consider the following query: $\pi_{until,type}(Rental \bowtie_{car=cid} Car)$

In this problem, you are asked to compute the communication cost (i.e., number of bytes transferred) for the given query and three different evaluation strategies. In all cases, to achieve the minimum communication cost, **project as early as possible to the attributes needed to evaluate the query**. `Rental.car` is a foreign key to `Car.cid`, hence *the join has a selectivity of* $\frac{1}{100}$. To keep things simple, we write kB as shorthand for 10^3 bytes.

Question 17: [1 credit] What is the communication cost if the query is evaluated at site 3?

(a) 306 kB **(b)** 406 kB **(c)** 506 kB **(d)** 406.5 kB **(e)** 506.4 kB **(f)** none **(Correct: a)**

Explanation:
Answer a is correct: We only need to send the join attributes + `until` + `type` to site 3, i.e.,
from the `Rental` relation, we have to send $10000 * (20 + 10) = 300$ kB
from the `Car` relation, we have to send $100 * (20 + 40) = 6$ kB
$\Rightarrow$ *total* $= 306$ kB

Question 18: [1 credit] What is the communication cost if the query is evaluated at site 1 and the result is transferred to site 3?

(a) 306 kB **(b)** 406 kB **(c)** 506 kB **(d)** 406.5 kB **(e)** 506.4 kB **(f)** none **(Correct: c)**

Explanation:
Answer c is correct:
from the `Car` relation, we have to send $100 * (20 + 40) = 6$ kB as before.
Due to the foreign key relationship, the join has 10,000 tuples, each of size $(10 + 40) = 50$ bytes, i.e., 500 kB
$\Rightarrow$ *total* $= 506$ kB

Question 19: [2 credits] Now suppose that the DDBMS first determines by a semi-join at site 2 those tuples of the `Car` relation that have a join partner. For this semi-join, you may assume selectivity = 0.8 (i.e., 80% of the tuples in the `Car` relation have a join partner; it also means that the `Rental` relation has only 80 distinct values in the `car` attribute).

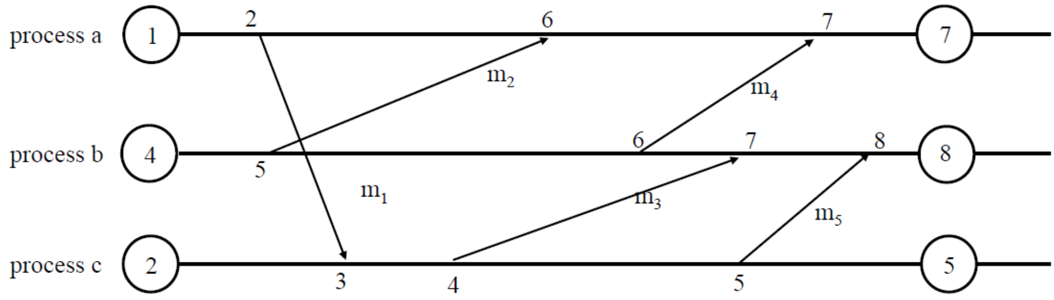
Here are the steps that the DDBMS carries out: it first sends the relevant information of the `Rental` relation to site 2 (assume duplicate elimination!) and determines by a semi-join with the `Car` relation those tuples of the `Car` relation that have a join partner. Then the DDBMS sends the relevant information of the `Car` relation to site 1 and computes the join result. Finally, the join result is transferred from site 1 to site 3. What is the total communication cost for this query plan?

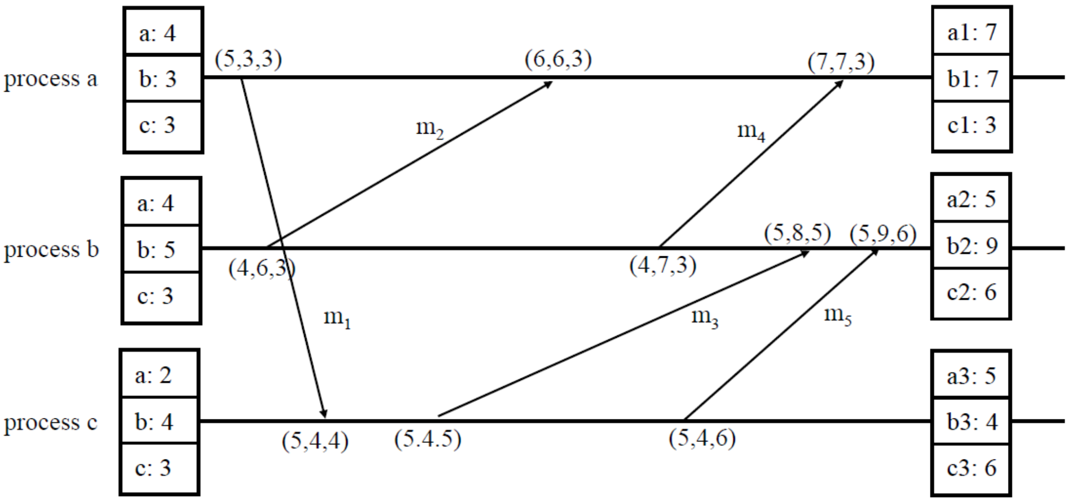
(a) 306 kB **(b)** 406 kB **(c)** 506 kB **(d)** 406.5 kB **(e)** 506.4 kB **(f)** none **(Correct: e)**

Explanation:
Answer e is correct:
transfer $\pi_{car}(Rental)$ to site 2: $80 * 20 = 1600 = 1.6$ kB
(notice the duplicate elimination: there are only 80 distinct values in attribute `car`)
transfer the result of the semi-join to site 1: $80 * (40 + 20) = 4800 = 4.8$ kB
then add the size of the join result: 500 kB
$\Rightarrow total = 506.4$ kB

Work sheet for notes. Do not hand in. Will not be graded.

Sample Solution of Problem 2





Sample Solution of Problem 3

Database DB1

```
Exams: [
{ ”student”: ”S1”,
”name”: ”alice”,
”studies”: 645,
”lecture”: ”L1”,
”title”: ”ADBS”,
”prof”: ”carol”,
”date”: 2024 },
{ ”student”: ”S2”,
”name”: ”bob”,
”studies”: 937,
”lecture”: ”L1”,
”title”: ”ADBS”,
”prof”: ”carol”,
”date”: 2025 },
{ ”student”: ”S1”,
”name”: ”alice”,
”studies”: 645,
”lecture”: ”L3”,
”title”: ”EP2”,
”prof”: ”emma”,
”date”: 2023 },
{ ”student”: ”S3”,
”name”: ”charlie”,
”studies”: 357,
”lecture”: ”L3”,
”title”: ”EP2”,
”prof”: ”emma”,
”date”: 2024 } ]
```

Database DB2

```
Students: [
{ ”s_id”: ”S1”,
”name”: ”alice”,
”studies”: 645,
”exams”:
  [ { ”l_id”: ”L1”,
  ”title”: ”ADBS”,
  ”prof”: ”carol”,
  ”date”: 2024 },
  { ”l_id”: ”L3”,
  ”title”: ”EP2”,
  ”prof”: ”emma”,
  ”date”: 2023 } ]
},
{ ”s_id”: ”S2”,
”title”: ”bob”,
”studies”: 937,
”exams”:
  [ { ”l_id”: ”L1”,
  ”title”: ”ADBS”,
  ”prof”: ”carol”,
  ”date”: 2025 } ]
},
{ ”s_id”: ”S3”,
”title”: ”charlie”,
”studies”: 357,
”exams”:
  [ { ”l_id”: ”L3”,
  ”title”: ”EP2”,
  ”prof”: ”emma”,
  ”date”: 2024 } ]
} ]
```

Database DB3

```
Students:
[ { ”s_id”: ”S1”,
”name”: ”alice”,
”studies”: 645,
”exams”: [”L1”,”L3”] },
{ ”s_id”: ”S2”,
”name”: ”bob”,
”studies”: 937,
”exams”: [”L1”] } ,
{ ”s_id”: ”S3”,
”name”: ”charlie”,
”studies”: 357,
”exams”: [”L3”] } ]

Lectures:
[ { ”l_id”: ”L1”,
”title”: ”ADBS”,
”prof”: ”carol”,
”exams”: [”S1”,”S2”] },
{ ”l_id”: ”L2”,
”title”: ”ThInf”,
”prof”: ”david”,
”exams”: [ ] },
{ ”l_id”: ”L3”,
”title”: ”EP2”,
”prof”: ”emma”,
”exams”: [”S1”, ”S3” ] } ]

Exams:
[ { ”student”: ”S1”,
”lecture”: ”L1”,
”date”: 2024 },
{ ”student”: ”S2”,
”lecture”: ”L1”,
”date”: 2025 },
{ ”student”: ”S1”,
”lecture”: ”L3”,
”date”: 2023 } ,
{ ”student”: ”S3”,
”lecture”: ”L3”,
”date”: 2024 } ]
```